JUnit5: Tagging and Filtering

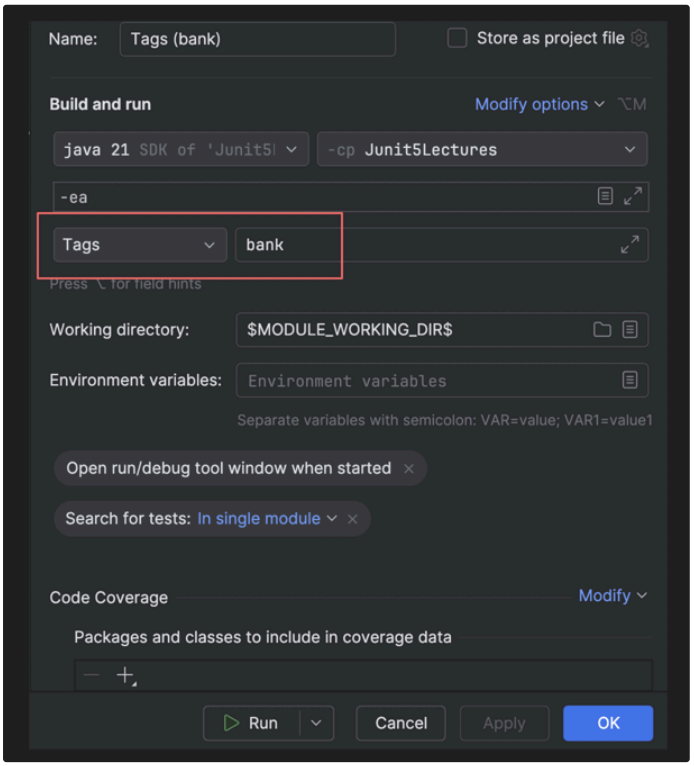
(Concept&&Coding - By Shrayansh)

Tagging:

Tags allow us to categorize tests, so that we can selectively run or skip them.

```java
class PaymentTaggingTest {

@Test
@Tag("card")
void validateCardScenario1() {
     System.out.println("inside Card testcase");
}

@Test
@Tag("bank")
void validateBankScenario1() {
     System.out.println("inside Bank testcase");
 }

}

```

Running the tests through IntelliJ:



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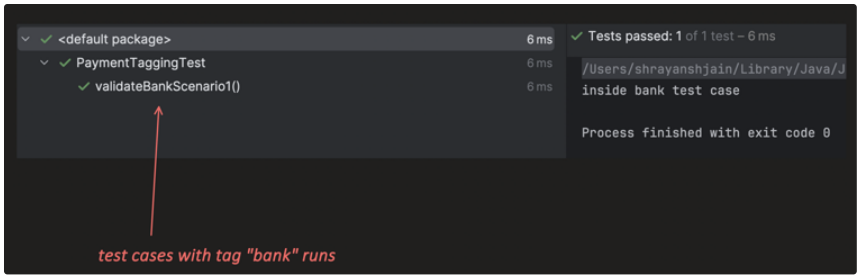

Syntax Rules for Tags:

1. **Tag must not be null or empty:** JUnit will ignore such tags.

```
@Test
@Tag(" ")
Void validateCardScenario1() {
     System.out.println("inside card test case");
}

```

2. **Stripped tag must not contain whitespaces:**

Internally Junit first *stripped* the tag, means leading and trailing whitespace is removed, after that there should not be any White Space.

```
//VALID
@Test
@Tag(" cardPayment ")
Void validateCardScenario1() {
     System.out.println("inside card test case");
}

//NOT VALID
@Test
@Tag("bank payment")
Void validateBankScenario1() {
     System.out.println("inside bank test case");
}


```

3. **Stripped tag must not contain any reserved characters:** JUnit explicitly bans these characters in tag names. Because below characters are used in Tag Expression (which is used for filtering)

,
(
)
&
|

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!

Multiple Tags and Class Tag:

```
//Class level tag, applied to all test methods
@Tag("slow")
class PaymentTaggingTest {

/*This test method has 3 tags associated:
    - card
    - visa
    - slow (class level one)
*/
@Test
@Tag("card")
@Tag("visa")
Void validateCardScenario1() {
     System.out.println("inside card test case");
}

/*
This test method has 2 tags associated:
    - bank
slow (class level one)
*/
@Test
@Tag("bank")
Void validateBankScenario1() {
     System.out.println("inside bank test case");
 }

}
```

Tag Expression:

- **Used for filtering the tags.**
- **It's a boolean expression with operators:**

,	(	)	&	|	!

Expression	Filter logic
card & visa	AND operator Test method should have both card and visa tag
card | visa	OR operator Any one tag should be present on Test method
!visa	NOT operator

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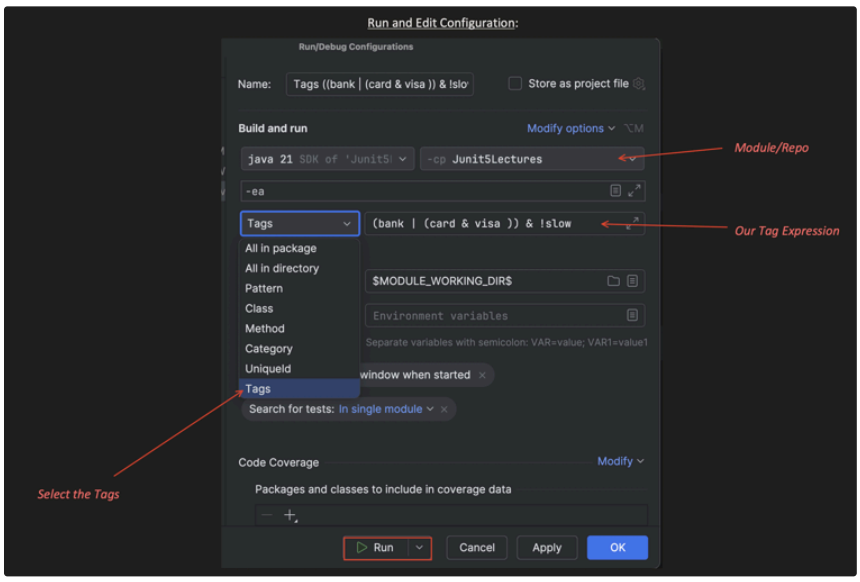
	All test which do not have visa tag
bank | ((card & visa) & !slow)	• Either bank • Or ◦ Card and visa tag should be there but it should not be slow

Execution:

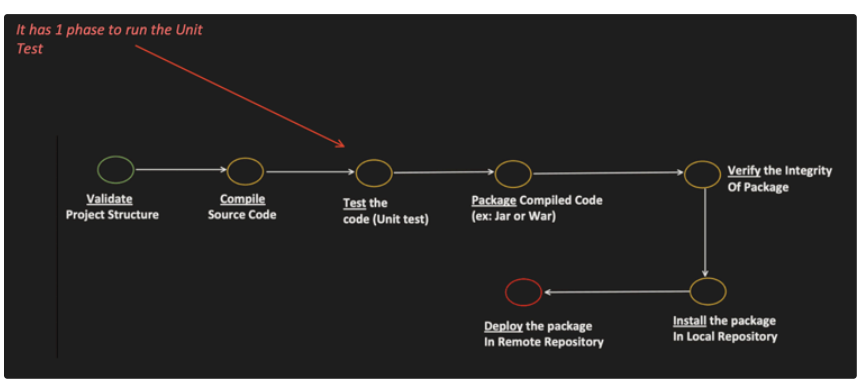
1. Through IntelliJ: IDE internally calls the same JUnit **Launcher API** and also pass the Tag info.



2. Through Maven Plugin:

We have already discussed Maven and its different lifecycle phases in Springboot series.
Pls check it out, if there is any doubt with Maven lifecycle.

Maven in-depth



pom.xml

```
```

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```
<build>
       <plugins>
              <plugin>
                     <artifactId>maven-surefire-plugin</artifactId>
                     <version>3.2.5</version>
                      <configuration>
                                 <groups>bank | card</groups>
                       </configuration>
              </plugin>
       </plugins>
</build>
```

Lets run the command: mvn test

```
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.662 s
[INFO] Finished at: 2025-11-19T02:07:10+05:30
[INFO] ------------------------------------------------------------------------
shrayanshjain@Shrayanshs-MacBook-Pro Junit5Lectures %
```

1 test for bank and 1 for card.

```
<build>
    <plugins>
        <plugin>
            <artifactId>maven-surefire-plugin</artifactId>
            <version>3.2.5</version>
             <configuration>
                  <groups>bank | card</groups>
             </configuration>
        </plugin>
    </plugins>
</build>
```

Maven auto adds it, but when we need to override the configuration, we need to add it explicitly.

But hardcoding like this is not recommended in production. So we use Profile.

pom.xml

```
<profiles>
       <profile>
           <id>tagTest</id>
           <build>
               <plugins>
                   <plugin>
                       <artifactId>maven-surefire-plugin</artifactId>
                       <configuration>
                           <!--suppress UnresolvedMavenProperty -->
                           <groups>${groups}</groups>
                       </configuration>
                   </plugin>
               </plugins>
           </build>
       </profile>
   </profiles>
```

Use maven profile with placeholder inside <groups>. This allow us to pass the tag at Runtime.

mvn test -PtagTest -Dgroups="bank | card"

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Profile is nothing but think of it as a SWITCH. Whenever we want to use it, we can activate it with -P<profile Name> like above.

And when we don't want to use it, we can skip it.

mvn test *this will run all the tests.*

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